Arjun Dahal

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Education

PhD in Computer Science — *University of Texas at Arlington*
Teaching Assistant Scholarship — *Aug 2022 – May 2027*

Bachelor's in Electronics and Communication Engineering — *IOE, Tribhuvan University*
Ranked in top 100 of 12,000 entrants, Full Scholarship — *Oct 2018*
Thesis: Nepali Speech Recognition using RNN-CTC Model

Experience

Graduate Teaching Assistant — *Arlington, TX*
University of Texas at Arlington — *Aug 2022 – Present*

- Conduct research on generating natural test cases for fairness testing and mitigating bias for ML models.
- Assist in teaching courses related to Programming, Software Development, and Software Testing.

Solutions Engineer — *Lalitpur, Nepal*
Guardsix (Formerly Logpoint) — *2018 – 2022*

- Solved software, system, and network issues in the production environment.
- Led the deployment and monitoring of SIEM/SOAR solutions across cloud and on-prem environments, developed custom analytics dashboards, alert pipelines, and event-driven orchestration scripts.
- Mentored new hires and led biweekly interdepartmental knowledge transfer sessions.

Research Projects

Bias Mitigation for Pretrained Classifiers — *2025 - Present*

- Developed a diagnostic-guided bias mitigation approach for pretrained tabular classifiers that applies minimal, dataset-adaptive changes to reduce the use of sensitive attribute information during the decision-making process.

Generation of Natural Test Cases for Fairness Testing — *2022 - 2025*

- Implemented a two-stage approach that either simply flips the sensitive attribute or uses an adversarial autoencoder for precise changes in related attributes for tabular instances. The generated instances are natural and high-proximity counterfactuals outperforming the state-of-the-art baselines in 8 out of 11 cases across 7 datasets.
- Applied Combinatorial Testing in the latent space of a Variational Autoencoder to efficiently generate natural test cases (improvement of 43% in naturalness over the baseline) for testing the fairness of black box models.

Nepali Speech Recognition — *2017 - 2018*

- Applied a Recurrent Neural Network along with Connectionist Temporal Classification (CTC) loss for end-to-end Nepali language recognition.

Publications

Dahal, A., Lei, Y., Kacker, R. N., & Kuhn, D. R. (2026). TabChange: An Approach to Natural Changes in Tabular Data. Under Review.

Dahal, A., Shree, S., Lei, Y., Kacker, R. N., & Kuhn, D. R. (2025). Fairness Testing of Machine Learning Models Using Combinatorial Testing in Latent Space. *In Proceedings of the 2025 IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*. IEEE.

Regmi, P., ***Dahal, A.***, & Joshi, B. (2019). Nepali Speech Recognition Using RNN-CTC Model. *International Journal of Computer Applications*, *178*(31), 1–6.

Skills

Programming: Python, C; Frameworks: Deep Learning (PyTorch), Machine Learning (Scikit-Learn), NumPy, Matplotlib; Domains: Tabular, NLP

Troubleshooting: System, Network, and Software debugging and troubleshooting

Leadership: IT Coordinator at Nepalese Student Association, University of Texas at Arlington (NSA-UTA)